

I2C Configuration Table

SM-A065M

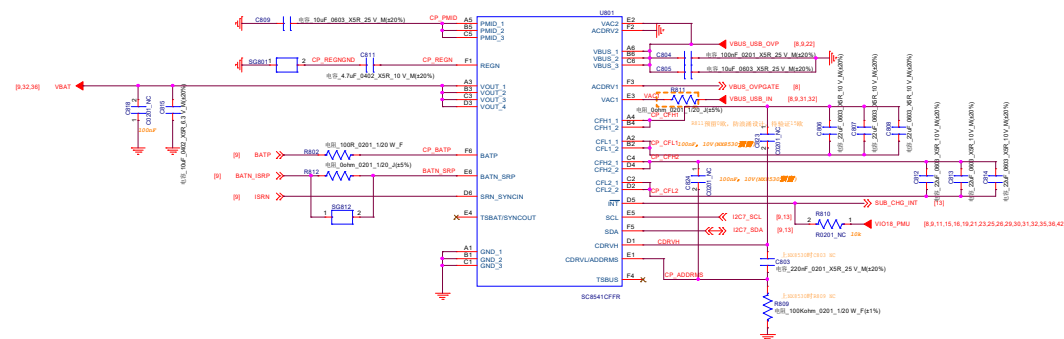
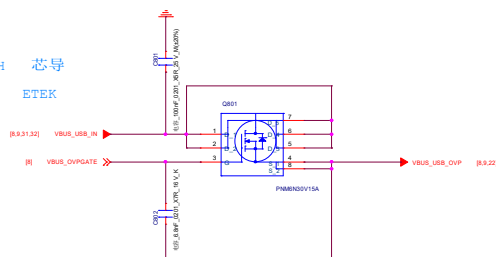
Function	I2C Slave Address (7-bit mode)
Page 15 : Charger IC	
Page 16 : LCM BIAS IC	
Page 21 : SMART PA	
Page 23 : Camera	
E-COMPASS	
ALS&PS SENSOR	
NFC	

华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title 02_I2C_ID_Overview		
Size D	Project A06_AL7160A	Rev v1
Date: Friday, May 24, 2024		Sheet 2 of 42

01.Block Diagram
02.I2C_ID_Overview
03.Index
04.BB_POWER_PDN
05.BB_POWER_IO
06.BB_1_GPIO
07.BB_2_GPIO
08.BB_3_EMI
09.BB_AUXADC
10.POWER_MT6357_Buck
11.POWER_MT6357_LDO
12.POWER_MT6357_IF
13.POWER_MT6357_Audio
14.POWER_BAT_CON
15.CHARGE_IC+CC
16.LCD BACKLIGHT+LCD BIAS
17.FLASHLED DRIVER
18.Memory_eMMC_LPDDR3
19.POWER_ThirdParty_Powers
20.Audio_EARJACK
21.Audio_MIC_Receiver
22.Audio_K8_PA
23.LCM_FPS
24.CAMERA
25.LINK
26.SIM_SD
27.SideKey_Flash_LED
28.Sensor
29.Test Point_JTAG
30.RF_MT6177_Pin_Out
31.RF_MT6177_RF_TX
32.RF_MT6177_RF_PRX
33.RF_MT6177_RF_DRX
34.RF_ANT
35.WCN_FEM
36.CONNECTIVITY_CONSYS_MT6631
37.SAR_Sensor

华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title 03_Index		
Size D	Project A06_AL7160A	Rev v1
Date: Friday, May 24, 2024		Sheet 3 of 42

HQ11230533000	PNM6N30V15H	芯導
TNM1230J	晶揚	
HQ163010005X0	EMR13N03C	ETEK

[illegible]

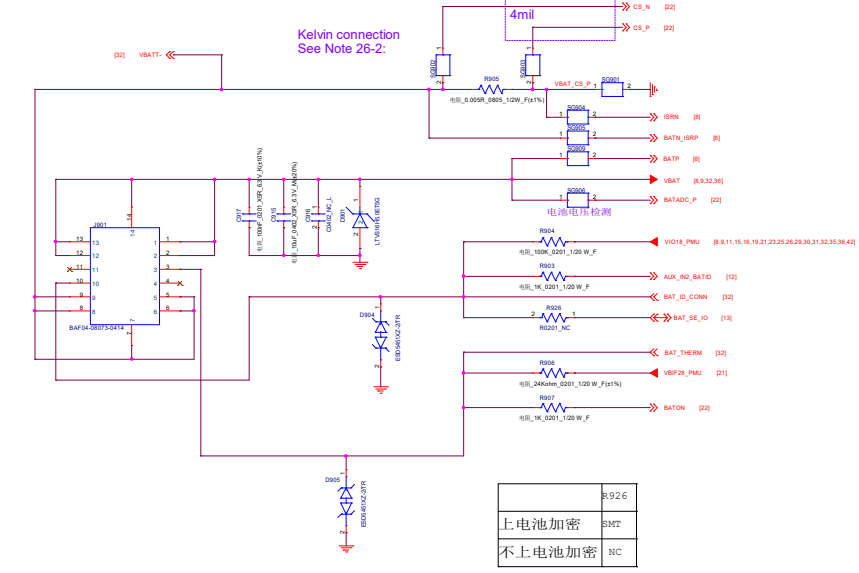
	U803	C822	R1203	R813
上USB防烧	SMT	SMT	NC	390
不上USB防烧	NC	NC	SMT	NC

备注：默认NC1导通，即检测FLASHLED NTC1

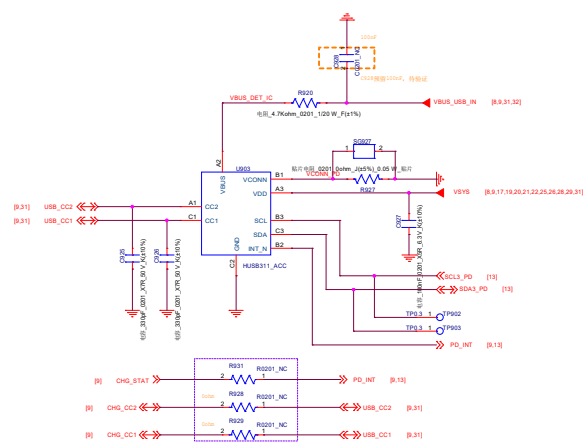
主供: HQ11155557000, DIO1520LP10, 帝奥微

华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title 08_Charge_Pump			
Size D	Project A06_AL7160A		Rev v1
Date: Friday, May 24, 2024		Sheet 8 of 42	

BATT CON BAT_ID/BAT_THERM上拉阻值及ADC采样范围需计算并确认 CS_P、CS_N 平行走线，宽度4mil，两边包地

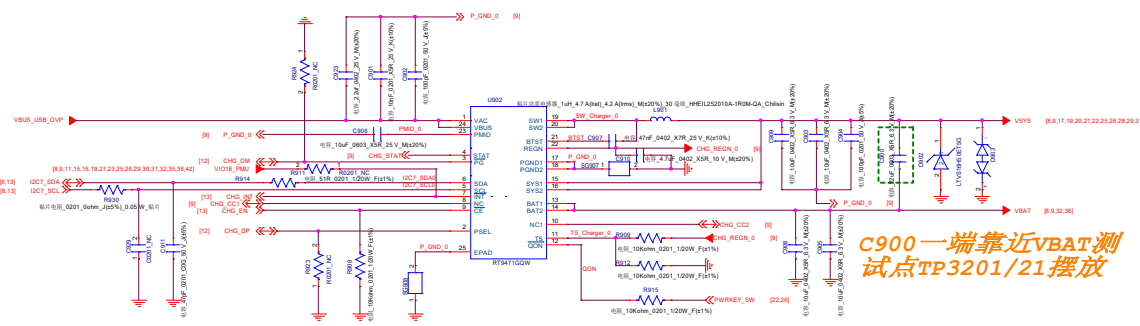


PD



兼容芯片集成式charger，贴芯片charger时，电阻贴0ohm

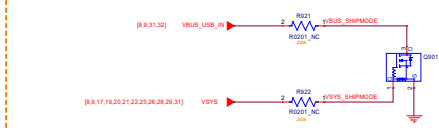
Charger



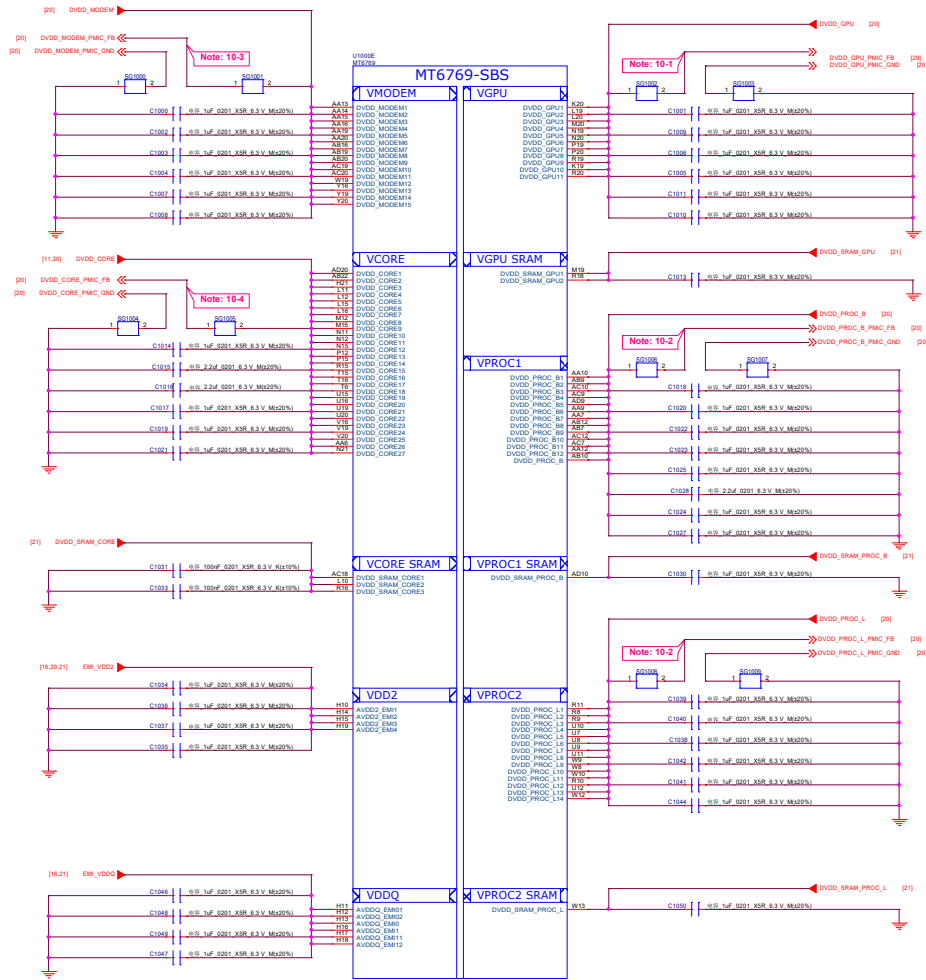
C900一端靠近VBAT测试点TP3201/21摆放

Ship Mode Det

默认NC



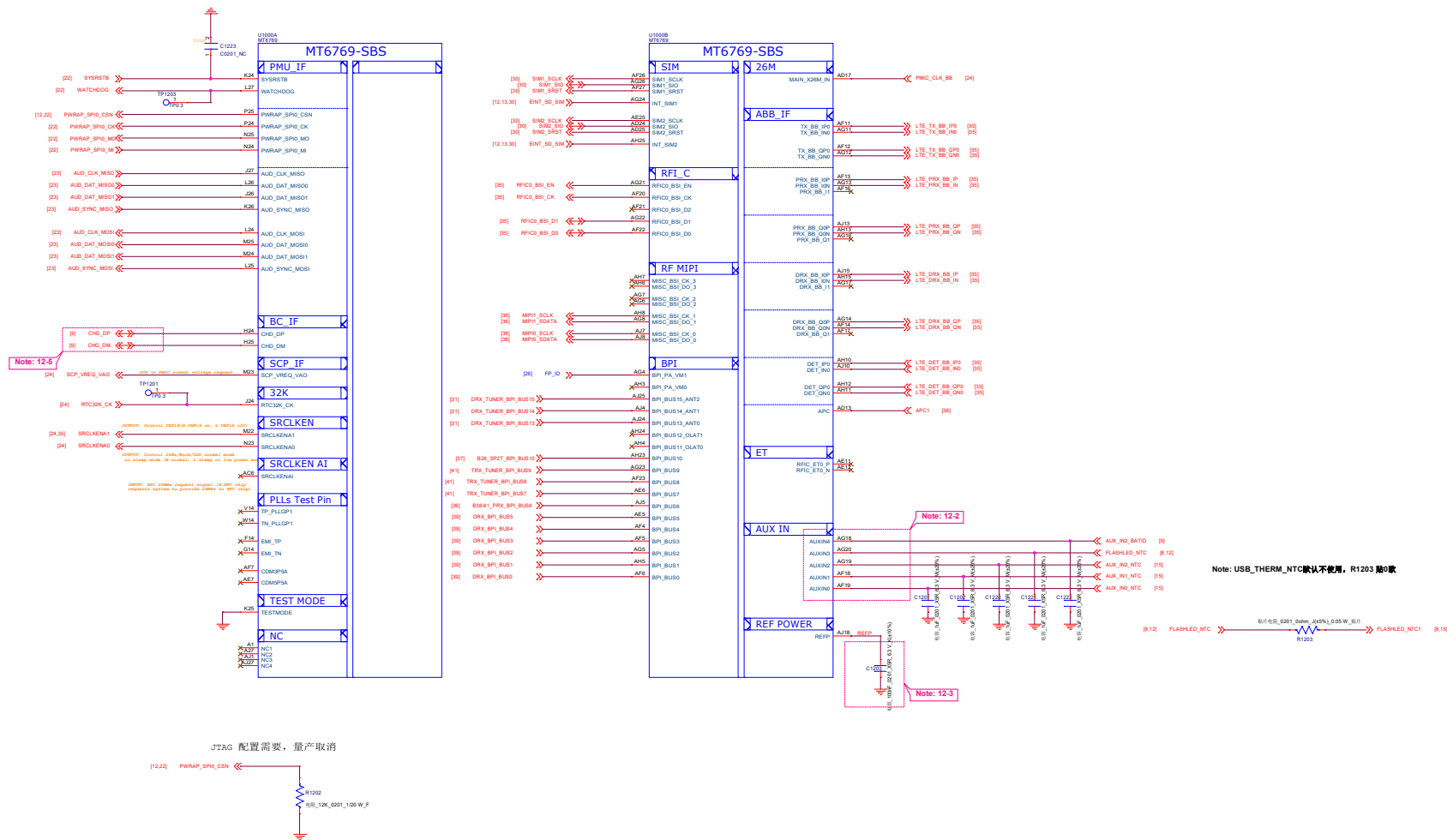
华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title 09_BAT_CHG_PD			
Size D	Project A06_AL7160A		Rev v1
Date: Friday, May 24, 2024		Sheet 9 of 42	



Schematic design notice of "10_BB_POWER_PDN" page:

- Note 10-1:** Differential pair of DVDD_GPU remote sense must be close to BB's ball.
Remote sense trace with GND shielding to PMIC (Differential)
- Note 10-2:** Differential pair of DVDD_PROC remote sense must be close to BB's ball.
Remote sense trace with GND shielding to PMIC (Differential)
- Note 10-3:** Differential pair of DVDD_MODEM remote sense must be close to BB's ball.
Remote sense trace with GND shielding to PMIC (Differential)
- Note 10-4:** Differential pair of DVDD_CORE remote sense must be close to BB's ball.
Remote sense trace with GND shielding to PMIC (Differential)

华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title 10_BB_POWER_PDN		
Size D	Project A06_AL7160A	Rev v1
Date: Friday, May 24, 2024	Sheet 10 of 42	



Schematic design notice of "12_BB_1" page:

Note 12-1: PWRAP_SPI0_CSN and "AUD_DAT_MOSIO" are bootstrap pins to select which interface will be the JTAG pin out

JTAG Function		JTAG Function	
PWRAP_SPI0_CSN	AUD_DAT_MOSIO	AP_JTAG	MD_JTAG
default=PU	default=PD	N/A	N/A
HI	LO	N/A	N/A
HI	HI (by ext. PU)	SPI0 + EINT8	SPI2 + SPI3
LO (by ext. PD)	LO	N/A	N/A
LO	HI	N/A	N/A

Note 12-2: To shunt a 1uF capacitor in the AUXIN ADC input to prevent noise coupling. It should be placed as close to BB as possible. Connect the unused AUX ADC input to GND.

Note 12-3: The de-coupling cap. for REFP (AJ18 ball) have to be placed as close to BB as possible.

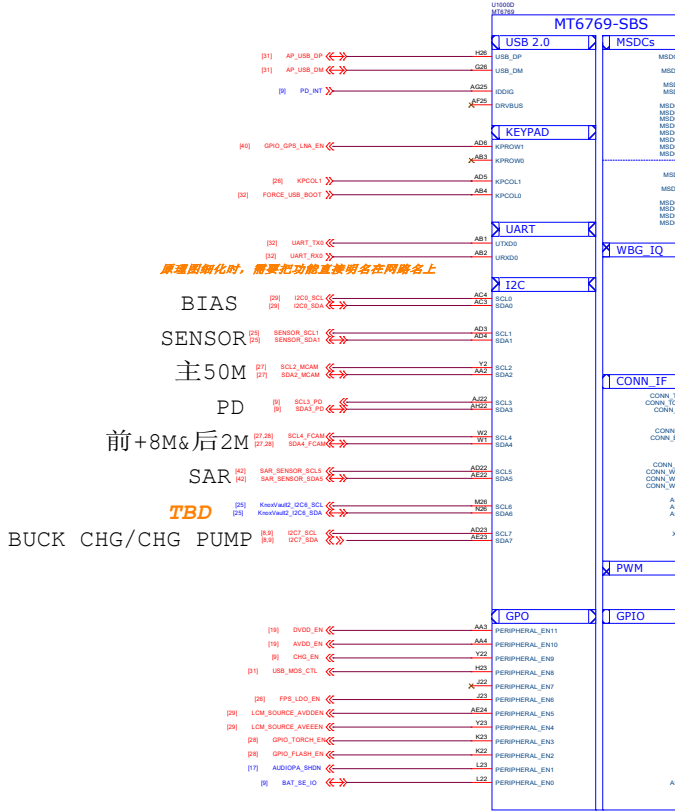
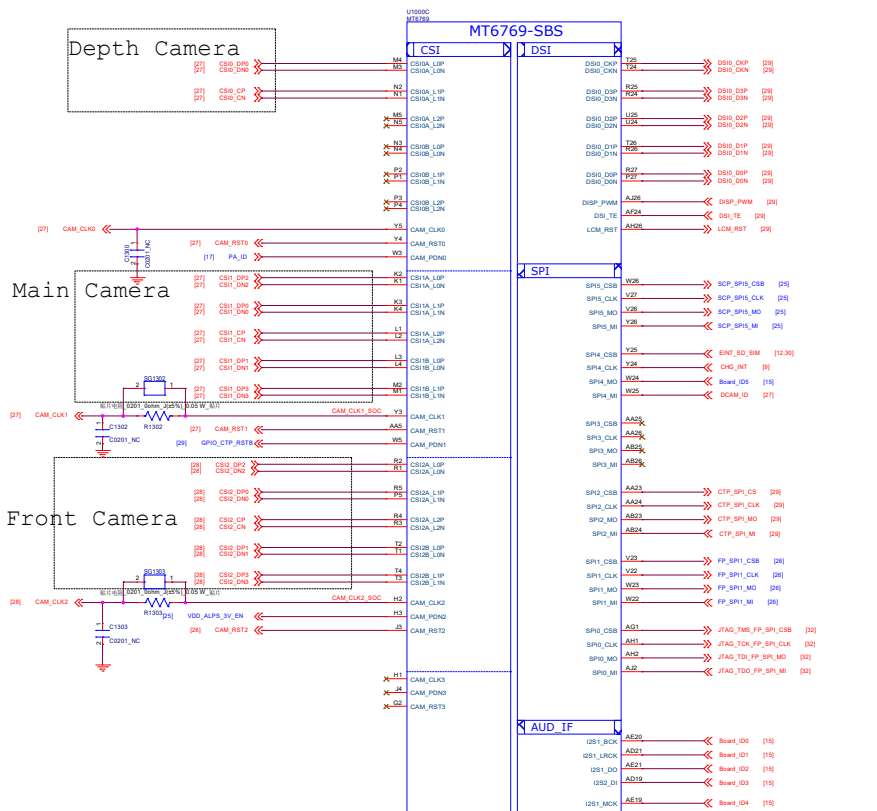
Note 12-4: HW pin for DDR type feature in bootstrap (refer to BB HW design Notice)

AUD_SYNC_MISO	AUD_CLK_MISO	CAM_PDN3	PMIC 6358 voltage	DDR Type
default=PD	default=PD	default=PD	VDRAM1 / VDRAM2	DDR
No ext. pull	No ext. pull	No ext. pull	1.125v / 0.6v	LP4X eMCP
No ext. pull	12K pull to VIO18	No ext. pull	OFF / 1.8v	Reserved
12K pull to VIO18	No ext. pull	No ext. pull	1.225v / 0.6v	LP3 eMCP
12K pull to VIO18	12K pull to VIO18	No ext. pull	1.125v / 1.8v	Reserved

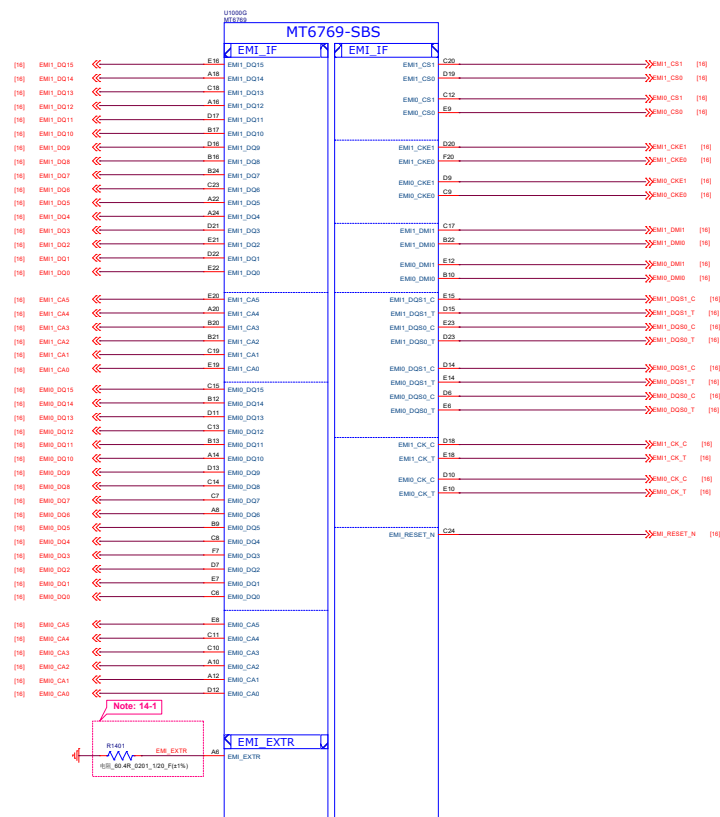
Note 12-5: Charger must have D+/D- pin for charger type USB detection. Charger must have at least 500mA USB current for All charger type.

华勤通讯 Huaqin Telecom Technology Com.,Ltd

Title			
12_BB_1			
Size	Project	Rev	
D	A06_AL7160A	v1	
Date:	Friday, May 24, 2024	Sheet	12 of 42



华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title 13_BB_2		
Size D	Project A06_AL7160A	Rev v1
Date: Friday, May 24, 2024		Sheet 13 of 42

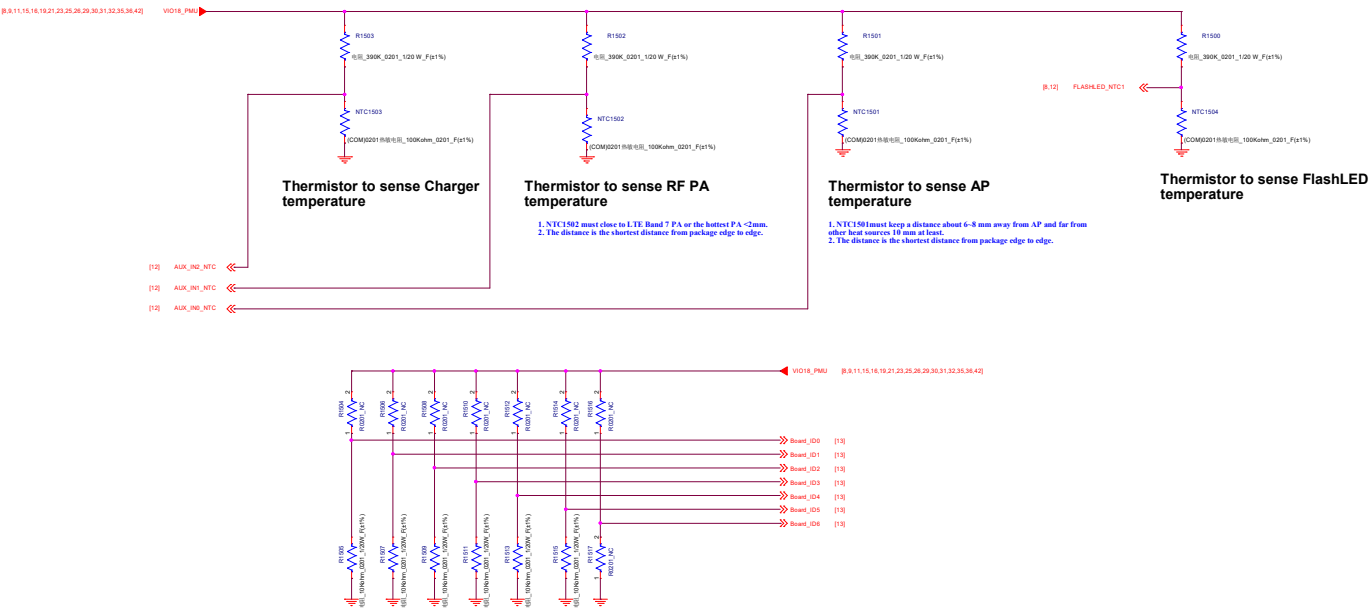


Schematic design notice of "14_BB_3" page.

Note 14-1: R1401 please select 60.4 ohm (1%) resistor

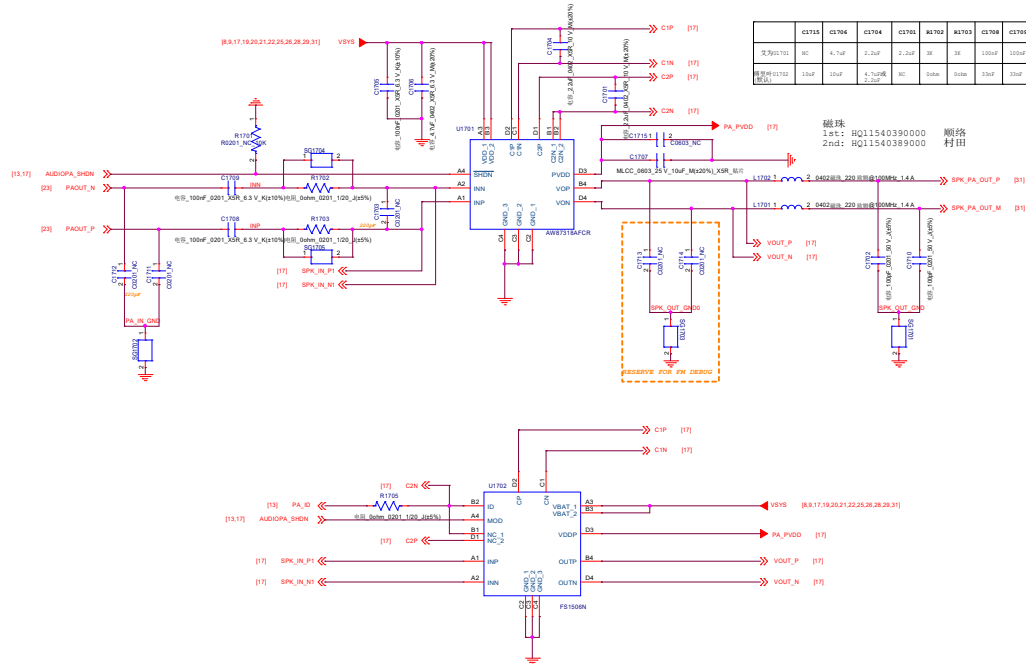
华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title 14_BB_3		
Size D	Project A06_AL7160A	Rev v1
Date: Friday, May 24, 2024		Sheet 14 of 42

Thermal/Board ID



华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title 15_BB_AUXADC_Thermal/Board ID		
Size D	Project A06_AL7160A	Rev v1
Date: Friday, May 24, 2024		Sheet 15 of 42

K类功放

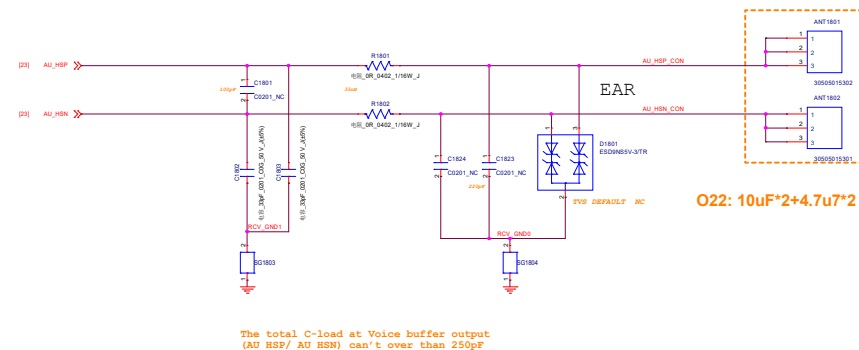


	C1715	C1706	C1704	C1701	R1702	R1703	C1708	C1709	R1705
艾为A1701	NC	4.7uF	2.2uF	2.2uF	3K	3K	100uF	100uF	NC
精英H1702 (默认)	10uF	10uF	4.7uF或 2.2uF	NC	0ohm	0ohm	33nF	33nF	NC

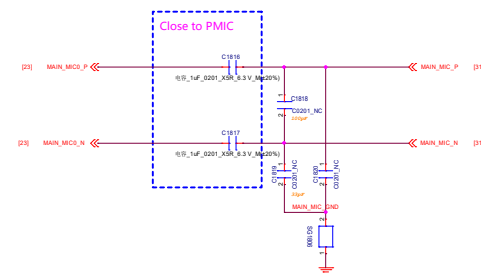
磁珠
1st: HQ11540390000 顺络
2nd: HQ11540389000 村田

华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title 17_AUDIO_PA		
Size D	Project A06_AL7160A	Rev v1
Date: Friday, May 24, 2024		Sheet 17 of 42

RECIPIENT



MAIN MIC KB

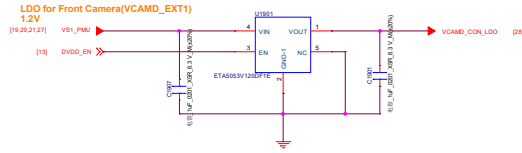


Title	18_MIC/RCV/Earphone
-------	---------------------

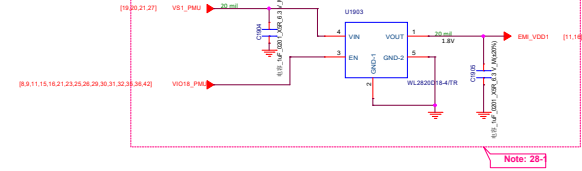
Rev
V1

Sheet 18 of 42

LDO for Front Camera(VCAMD_EXT1) 1.2V

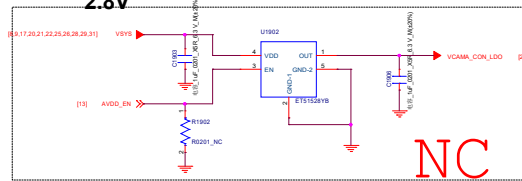


Ext. EMI_VDD1 for LPDDR4X VDD1



U1903
主供: HQ1152534000, ETA5055V180DF1E, LDO_1.8V_500 mA, ETA (钰泰)
二供: 暂无料号, WR0516-18FF4R, 维安
三供: HQ11555CJ000, ET55618Y1B, LDO_1.8_500 mA_ET55618Y1B, ETEK (力芯微)

LDO for 50M Camera (VCAMA_EXT) 2.8V



AVDD 兼容LDO
使用平台 LDO NC

Schematic design notice of "28_POWER_ThirdParty_Powers" page:

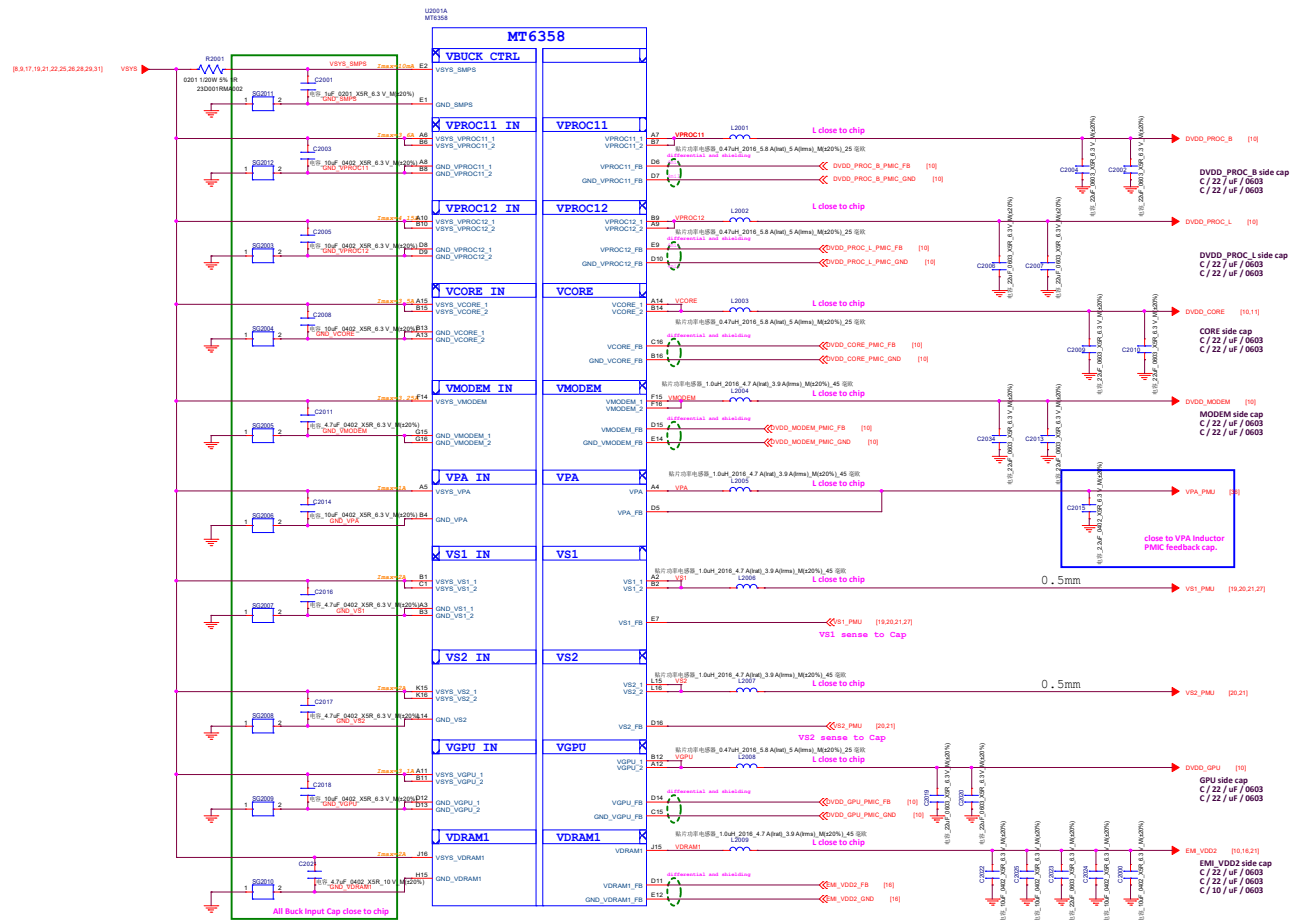
Note 28-1: Refer to BB design notice for LP4X VDD1 LDO selection guide

华勤通讯 Huaqin Telecom Technology Com.,Ltd

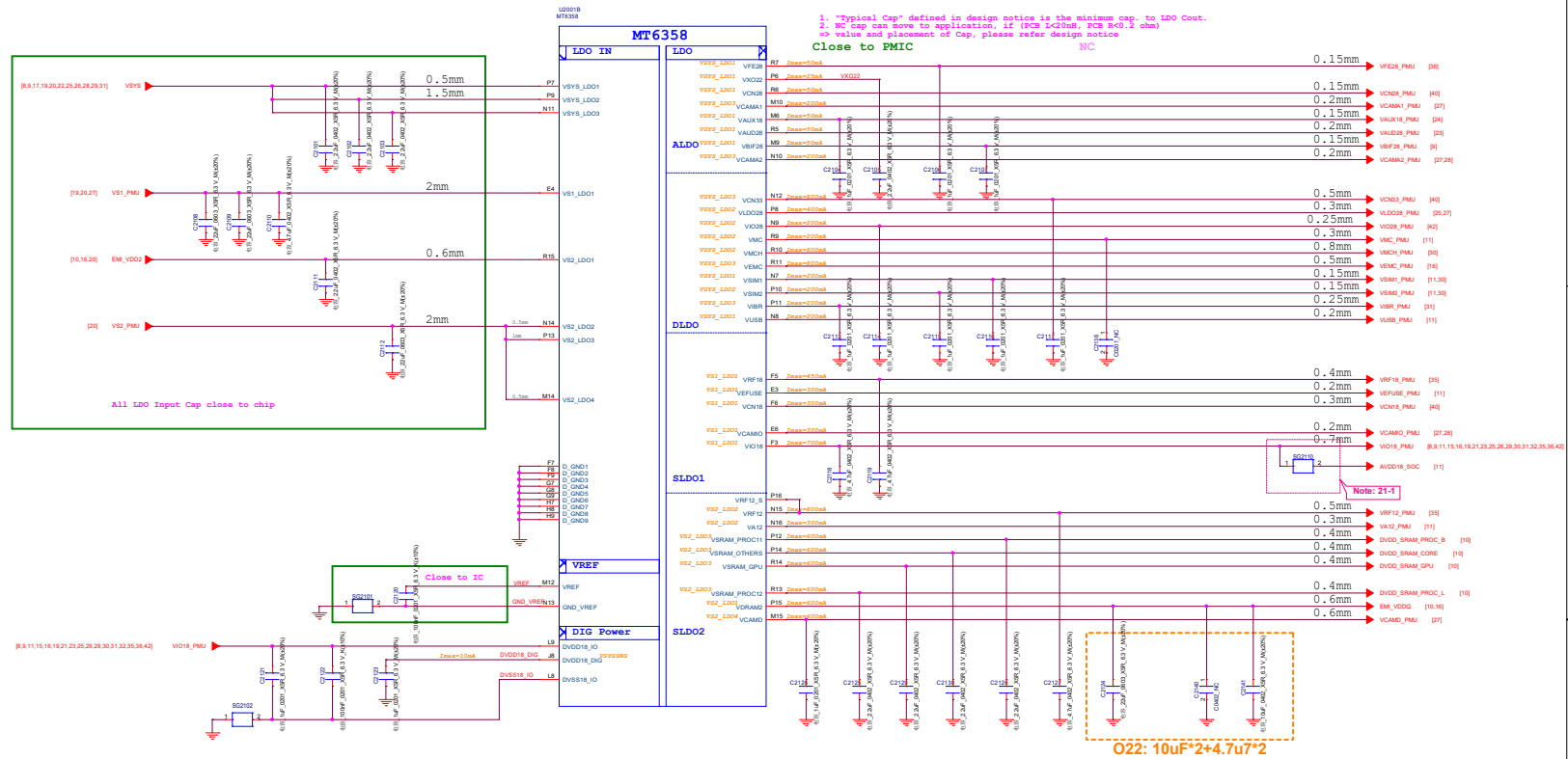
Title A19_POWER_ThirdParty_Powers

Size D	Project A06_AL7160A	Rev v1
--------	---------------------	--------

Date: Friday, May 24, 2024	Sheet 19 of 42
----------------------------	----------------



华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		20_POWER_MT6358_BUCK	
Size	Project	A06_AL7160A	
D		v1	
Date: Friday, May 24, 2024		Sheet	20 of 42



Schematic design notice of "21_POWER_MT6358-LDO" page.

Note 21-1: Please set SG2110 close to C2118, making star connection between VIO18_PMU and AVDD18_SOC near to LDO cap. C2118
 Please also refer to MT6358 design notice for further detail design information

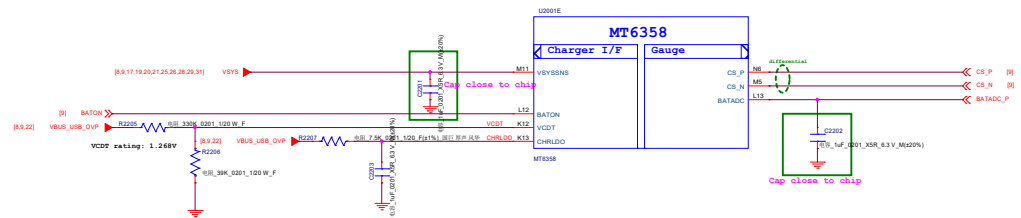
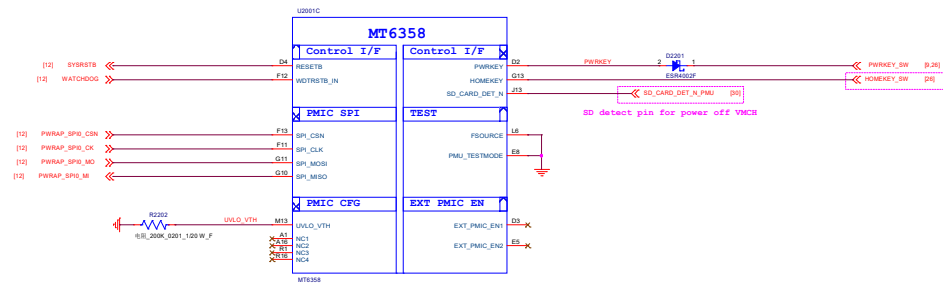
Note 21-2: If these power trace can meet LDO layout constraint, these CAP can be NC or removed.
 Please refer to MT6358 design notice.

华勤通讯 Huaqin Telecom Technology Com.,Ltd

Title
 21_POWER_MT6358_LDO

Size D	Project A06_AL7160A	Rev v1
-----------	------------------------	-----------

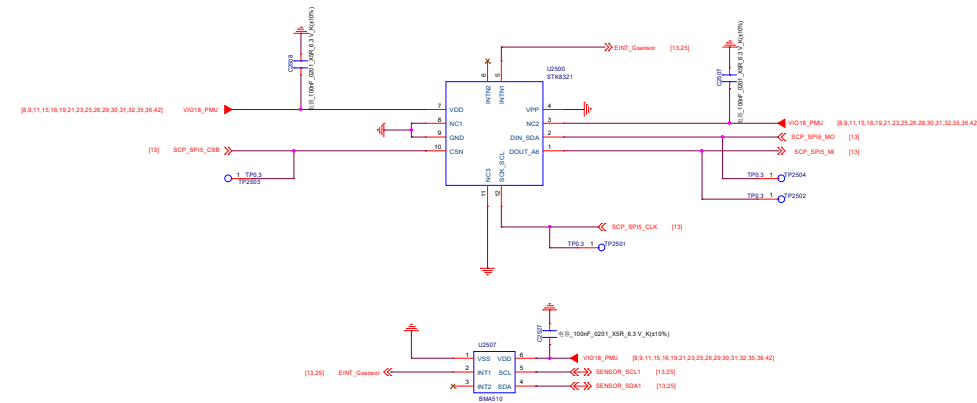
Date: Friday, May 24, 2024 Sheet 21 of 42



华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title 22_POWER_MT6358_IF		
Size D	Project A06_AL7160A	Rev V1
Date: Friday, May 24, 2024		Sheet 22 of 42

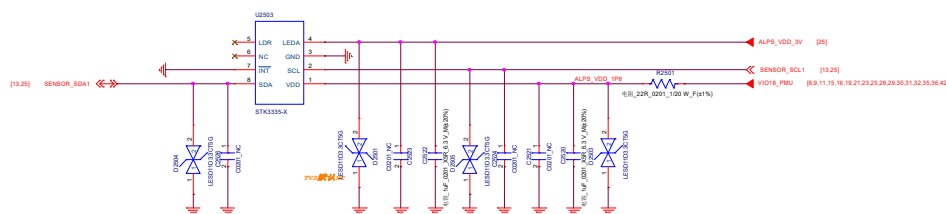
eCOMPASS

G SENSOR

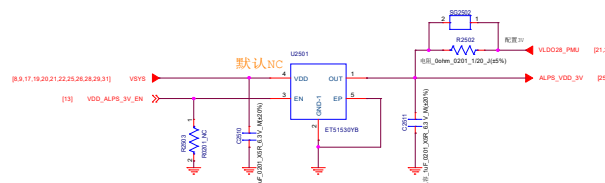


U2507为小封装兼容方案，
若贴U2507，则U2500及所有外围均NC
若不贴U2507，则U2500及外围器件全部上件

ALPS Sensor



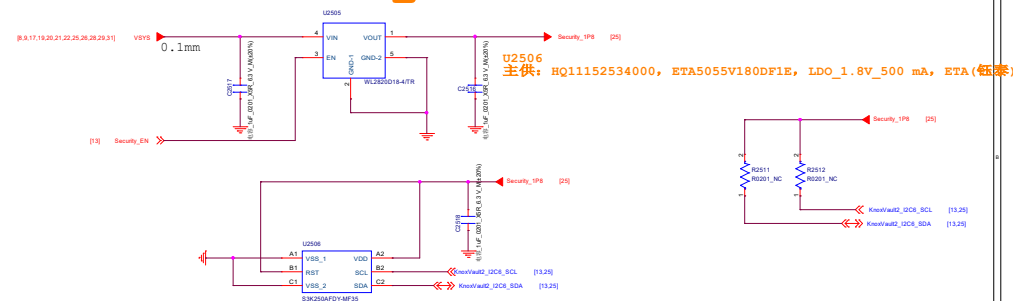
LDO For ALP



LDO For ALP, 设置平台电压3V, 若有问题, 则上3V LDO

	R2502	U2501	C2510	C2511
平台供电 (默认)	SMT	NC	NC	NC
LDO供电	NC	SMT	SMT	SMT

Security IC



U2506
主供: HQ133010000Y0, S3K250AFDY-MF35, Samsung (三星电子)

华勤通讯 Huaqin Telecom Technology Com.,Ltd

Title 25_SENSOR

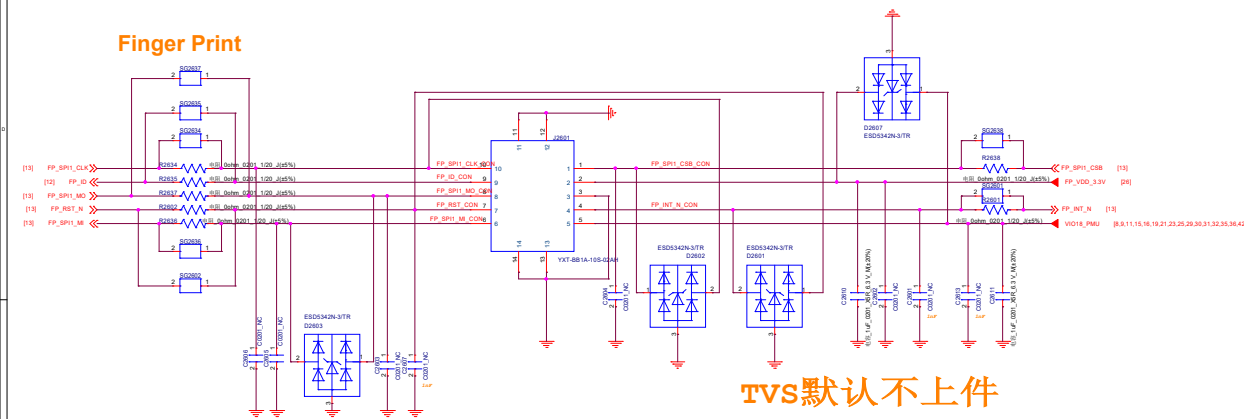
Size D Project A06_AL7160A

Rev v1

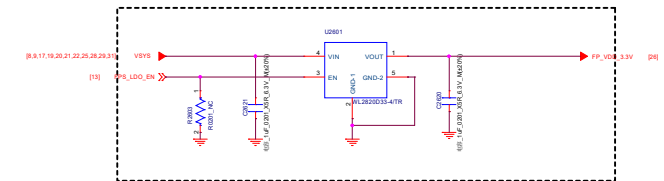
Date: Friday, May 24, 2024

Sheet 25 of 42

Finger Print



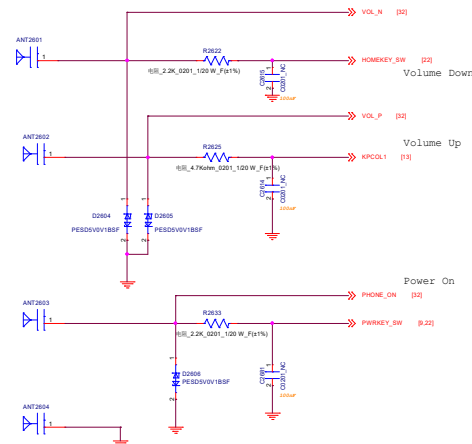
LDO for FPS



主供: HQ11152105000, ETA5053V330DF1E,
二供: HQ11153588000, WL2820D33-4/TR,
三供: HQ11153571000, AW37030D330DNR,

TVS默认不上件

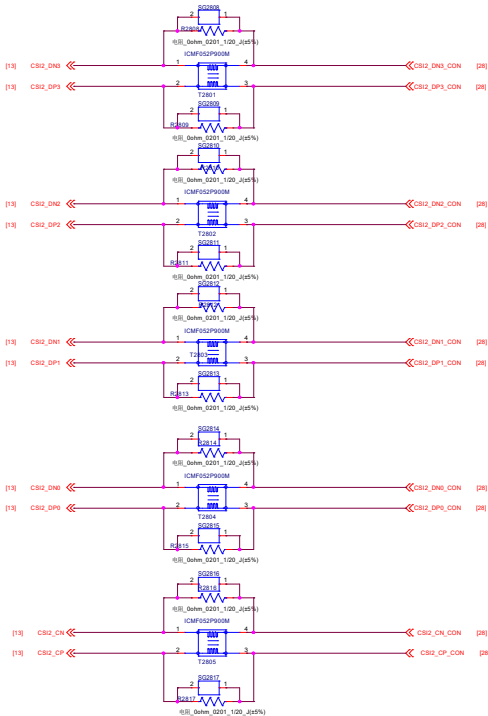
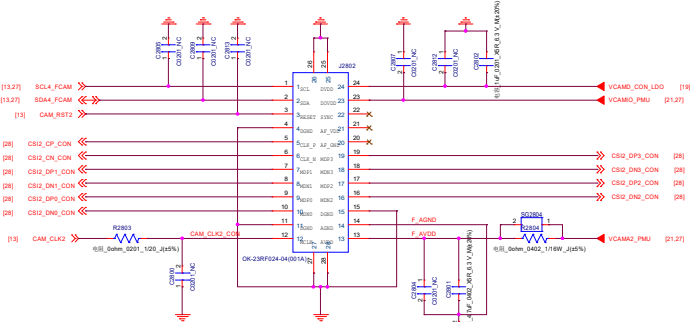
KEY pad



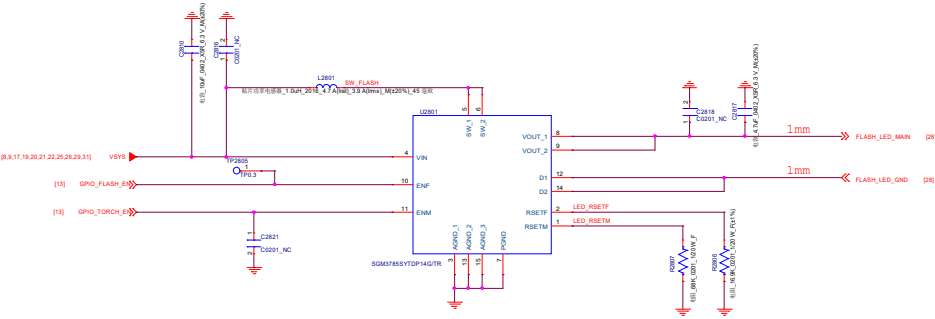
华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title 26_FP/KEY		
Size D	Project A06_AL7160A	Rev v1
Date: Friday, May 24, 2024		Sheet 26 of 42

Front 8M Cam

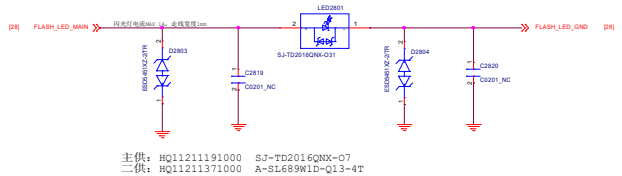
Front Camera电源走线要求:
AVDD 0.2mm
DVDD 0.3mm
IOVDD 0.1mm



Flash Driver



Flash LED

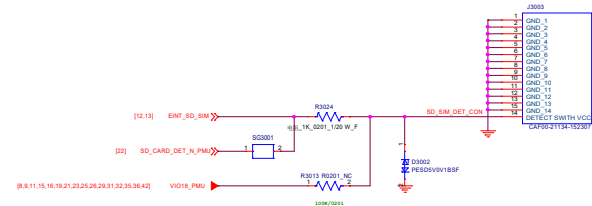
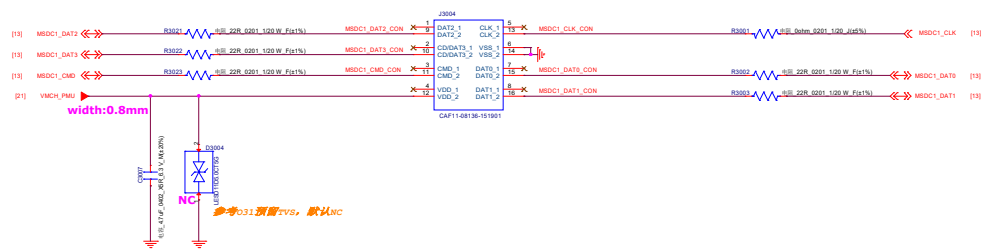


替代关系	料号	型号	供应商	R2807	R2806	If1ash	Itorch	封装类型	封装尺寸(MAX)
主供	HQ111555CH000	SQM37858VTD014G/TP	SGS (意邦微)	68K	12K	1500mA	200mA	TDPN-3*2-14L	3.1*2.1*0.8mm
二供	HQ11154488000	AS658580BNC	Amphenol (昂赛微)	68K	12K	1500mA	800mA	DPN2030-14L	3.1*2.1*0.8mm
三供		AM36501	艾为						

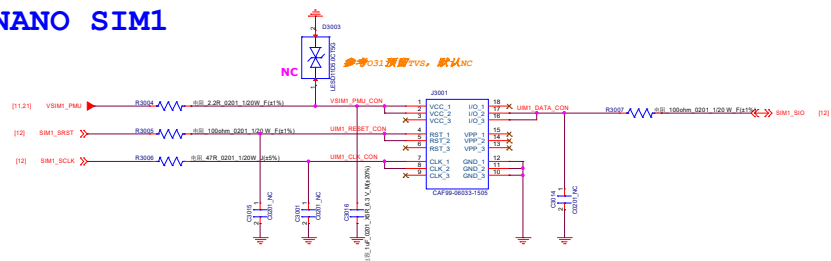
华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title 28_FRONT_CAM/FLASH		
Size D	Project A06_AL7160A	Rev v1
Date: Friday, May 24, 2024		Sheet 28 of 42

3选3卡座

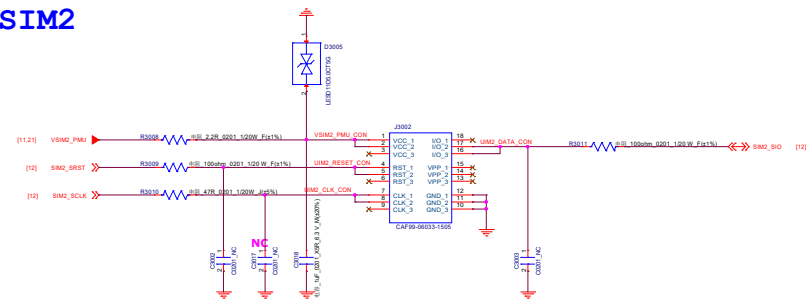
TF CARD



NANO SIM1



NANO SIM2



华勤通讯 Huaqin Telecom Technology Com.,Ltd

Title	30_SIM&TF
-------	-----------

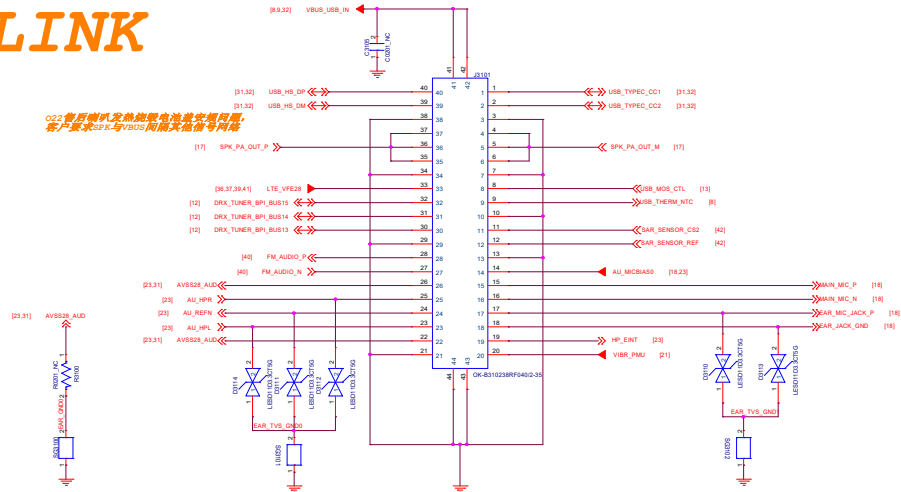
Size	Project	A06_AL7160A
D		

Rev
v1

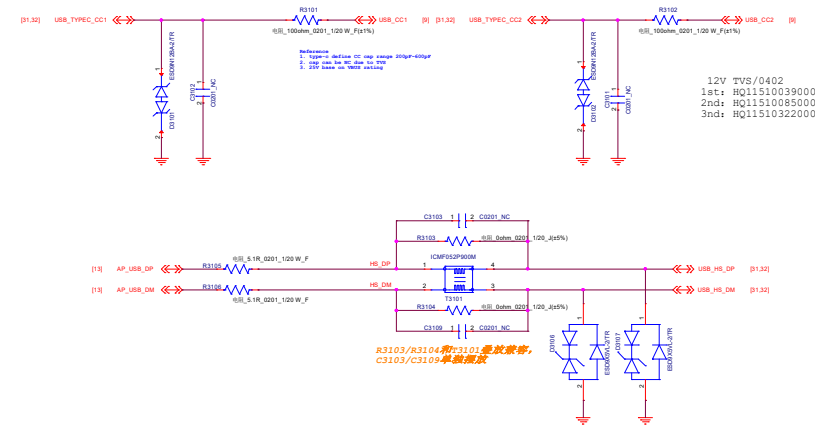
Date: Friday, May 24, 2024

Sheet 30 of 42

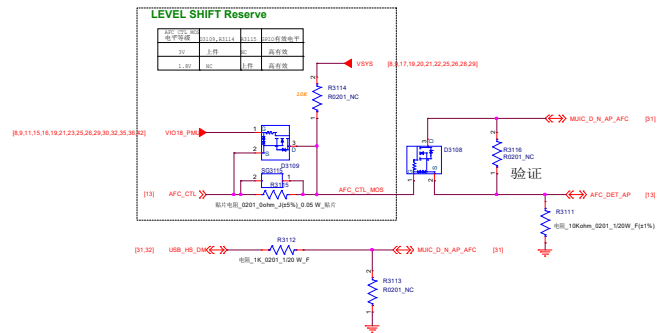
LINK



CC&DP/DM



AFC



OVP REPLACED BY NMOS

华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title 31_LINK&CC&DPDM&AFC			
Size D	Project A06_AL7160A		Rev v1
Date: Friday, May 24, 2024		Sheet 31 of 42	



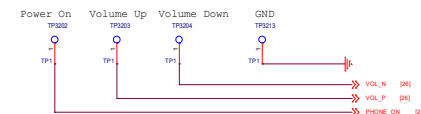
Shielding GND



MARK



KEYPAD



JTAG Connector



华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title 32_TEST_POINT/SHIELD/JTAG			
Size D	Project A06_AL7160A		Rev v1
Date: Friday, May 24, 2024		Sheet 32 of 42	

华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title 33_BB_RESERVE		
Size D	Project A06_AL7160A	Rev v1
Date: Friday, May 24, 2024		Sheet 33 of 42

华勤通讯 Huaqin Telecom Technology Com.,Ltd

Title 34_NFC

Size D	Project A06_PA9540	Rev v1
-----------	-----------------------	-----------

Date: Friday, May 24, 2024	Sheet 34 of 42
----------------------------	----------------

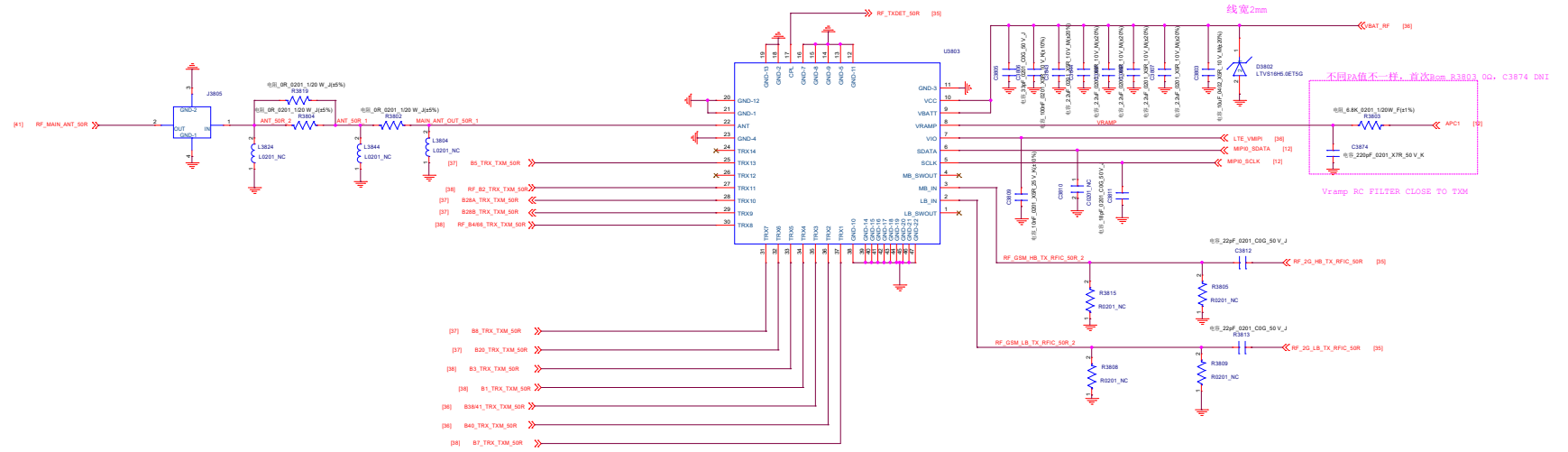
**LATAM B13
GLOBAL/EUR B28B**

The schematic diagram illustrates the RF front-end of the LATAM B13 GLOBAL/EUR B28B. It features three main RF modules: B28B_PX1, B28B_PX2, and B28B_PX3. The B28B_PX1 module is connected to the TX and RX paths, with its TX output going to the antenna and its RX input receiving signals from the B28B_PX2 and B28B_PX3 modules. The B28B_PX2 module is connected to the TX path, with its TX output going to the antenna and its RX input receiving signals from the B28B_PX3 module. The B28B_PX3 module is connected to the RX path, with its RX input receiving signals from the B28B_PX2 and B28B_PX1 modules. The diagram shows the matching networks, filters, and the final output to the antenna. Key components include the B28B_PX1, B28B_PX2, and B28B_PX3 modules, the B28B_PX1 and B28B_PX2 filters, the B28B_PX3 filter, the B28B_PX1 and B28B_PX2 matching networks, and the B28B_PX3 matching network. The diagram also shows the power supply connections for the modules, including the 1.8V and 3.3V rails. The layout is labeled with component values and part numbers, and includes a note 'Close to MT6177' indicating the proximity to the main processor.

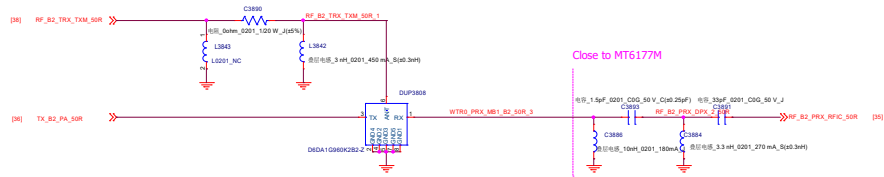
[illegible]

Date: Friday, May 24, 2024 Sheet 37 of 42

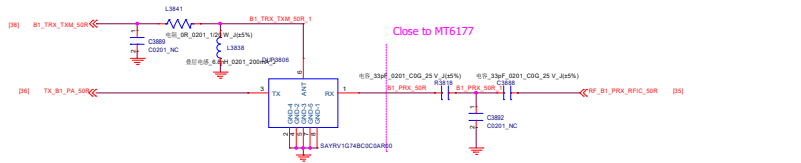
Main ANT



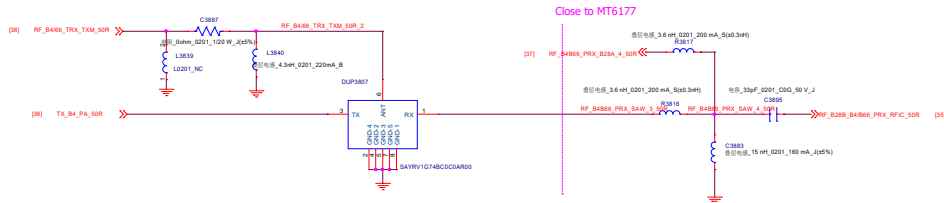
B2 TRX



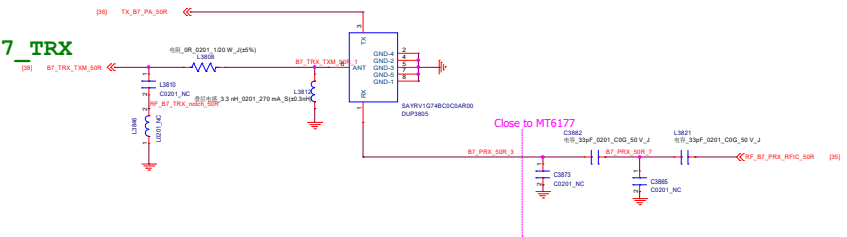
B1 TRX



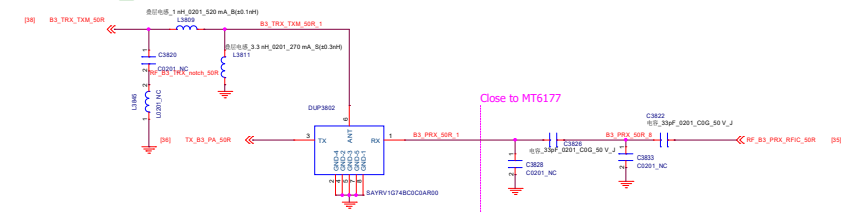
B4/66 DPX



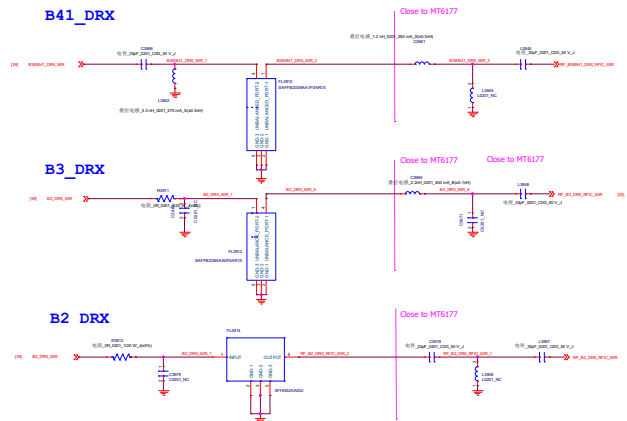
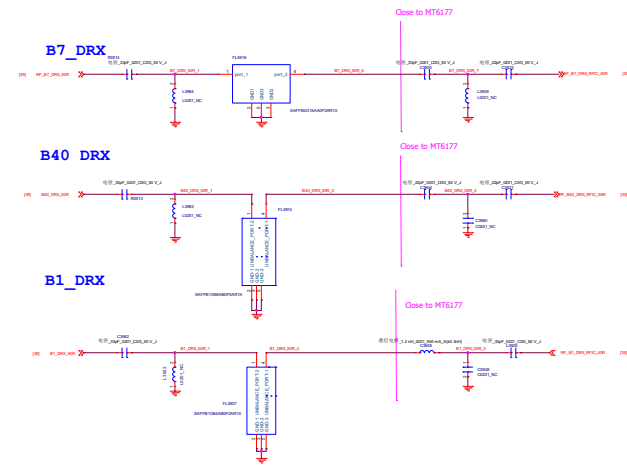
B7 _TRX



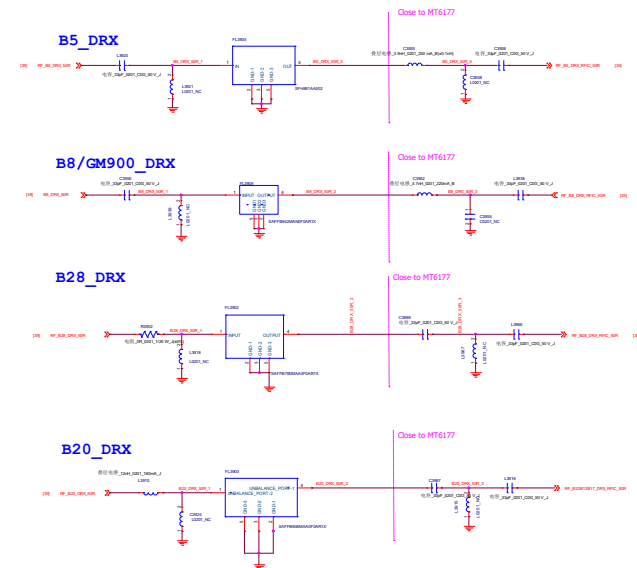
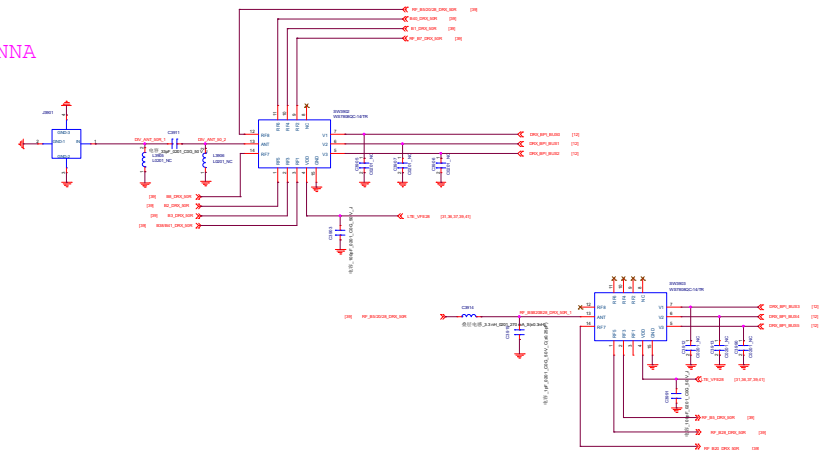
B3_TRX



华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title 38_RF_TXM			
Size D	Project A06_PA9540		Rev V1
Date: Friday, May 24, 2024		Sheet 38 of 42	



DRX_ANTENNA



华勤通讯 Huaqin Telecom Technology Com.,Ltd

Title 39_DRX

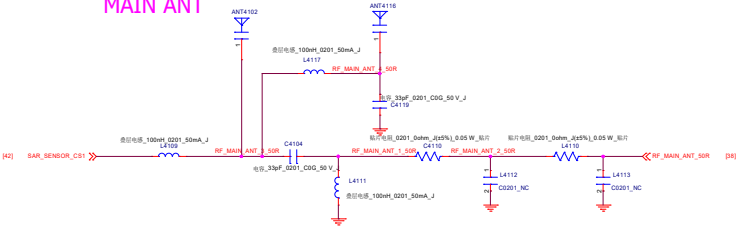
Size E Project A06_PA9540

Rev v1

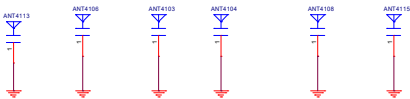
Date: Friday, May 24, 2024

Sheet 39 of 42

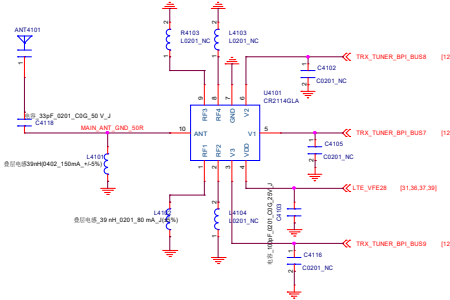
MAIN ANT



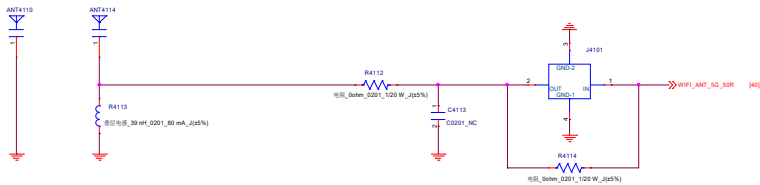
TOP面接地弹片



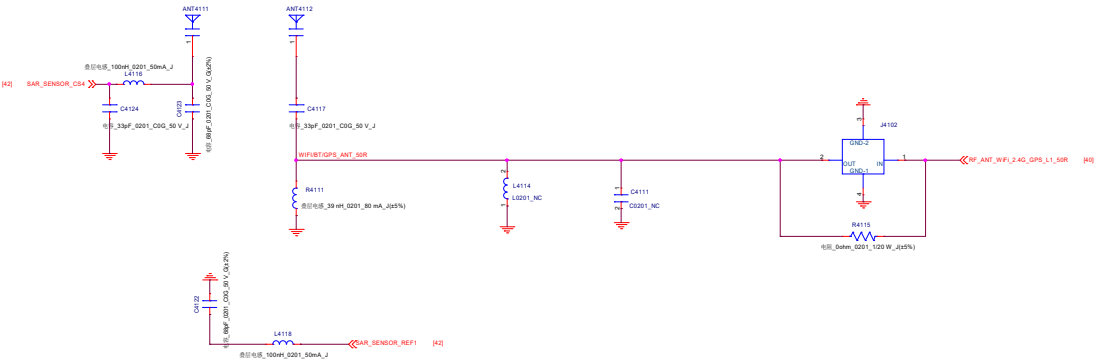
TUNER



Wifi 5G ANT

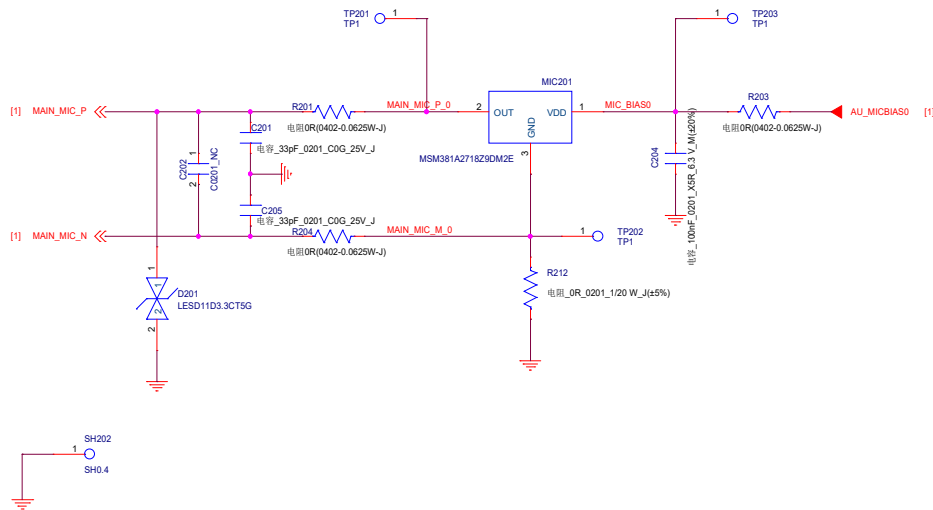


2.4G_GPS ANT

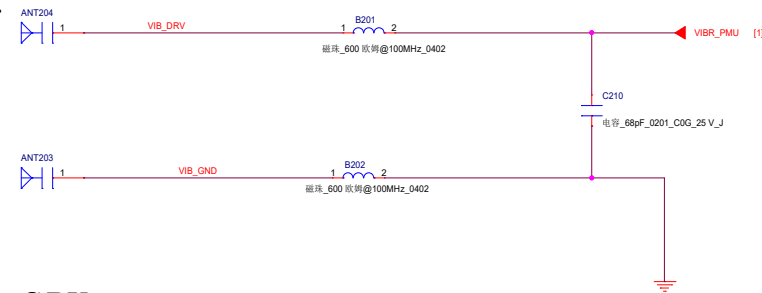


华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title 41_ANT		
Size D	Project A06_PA9540	Rev v1
Date: Friday, May 24, 2024		Sheet 41 of 42

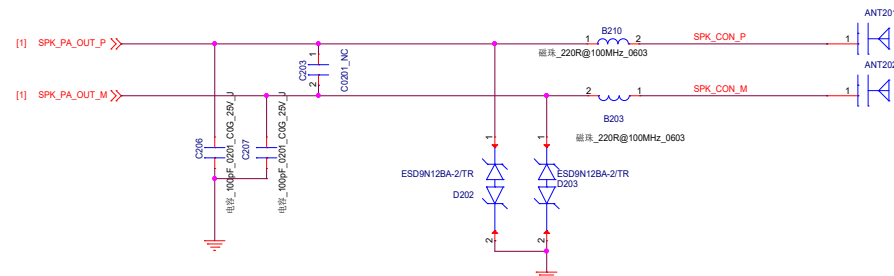
MIC



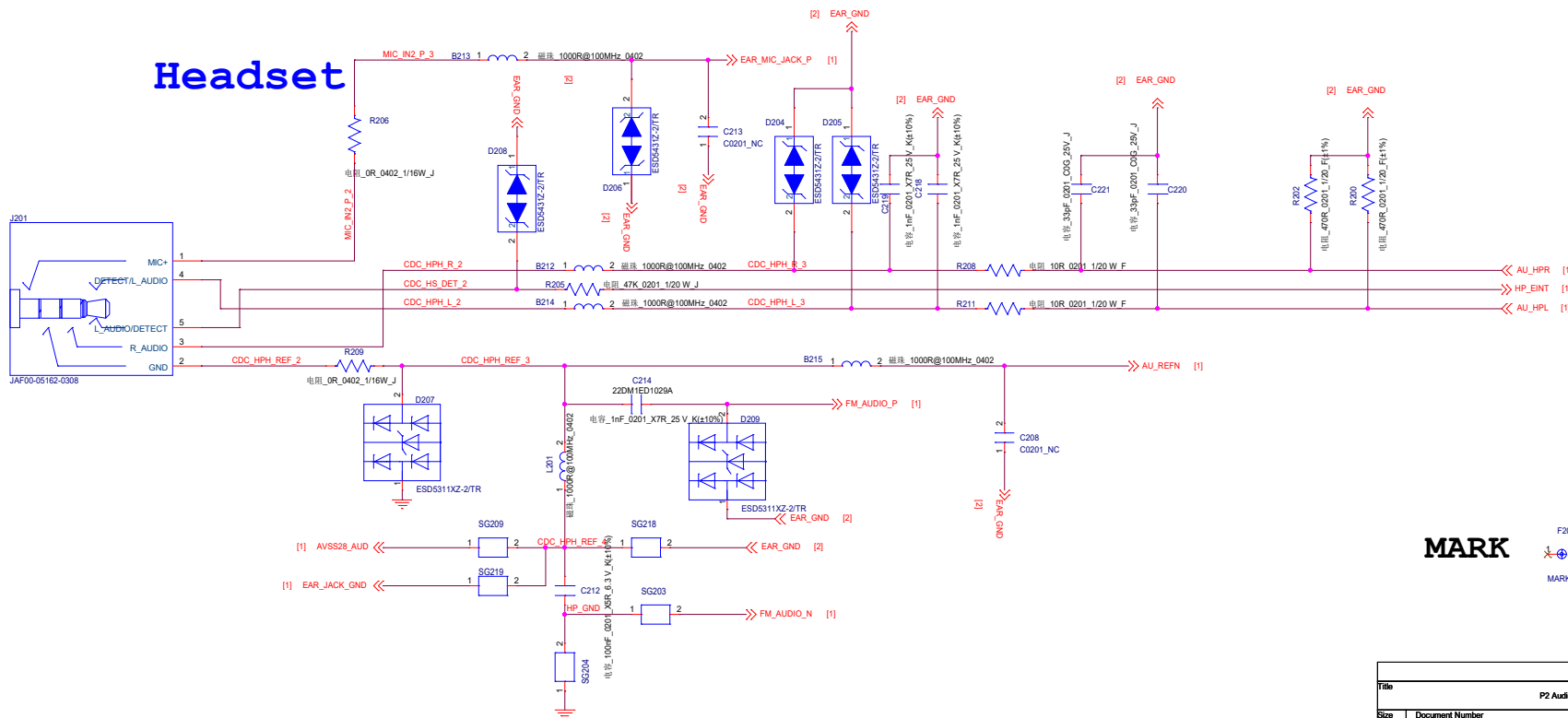
MOTOR



SPK



Headset

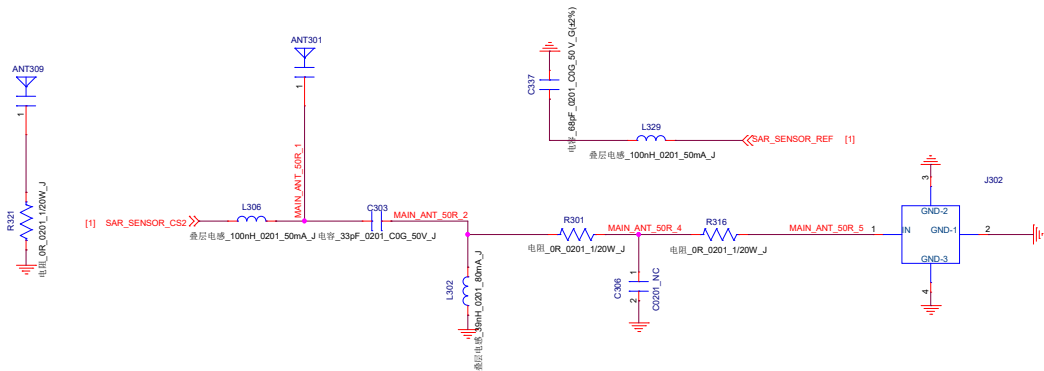


MARK

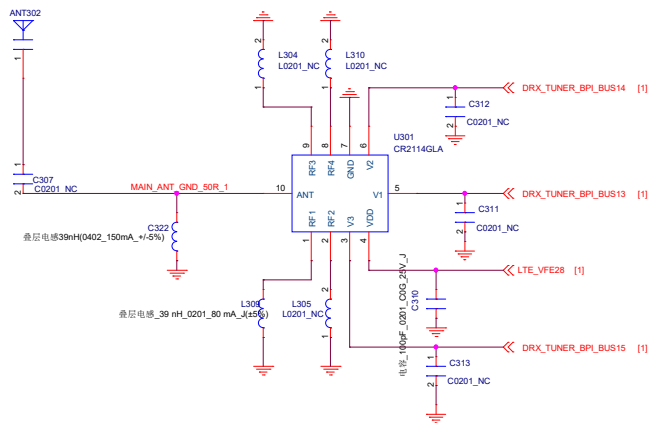


Title			P2 Audio
Size	Document Number	A06_AL7160A	
C		Rev	
Date			Wednesday, March 13, 2024
Sheet			2 of 3

D

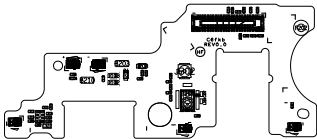


C



B

A



SM-A065M